

ThreatLens

Turning Noisy Security Alerts into Correlated, Prioritized & Actionable Incidents

Team: Threat Minds

Hackathon: IBM BoB AI Innovation Hackathon 2026

Domain: Cybersecurity / AI-assisted Security Operations

AI-assisted security monitoring and incident response platform that helps security analysts move from raw multi-source security alerts to correlated, explainable, and prioritized security incidents.

25

Security Alerts

Raw inputs ingested from multiple sources

8

Correlated Incidents

Intelligently grouped and deduplicated

95

Critical Risk Score

Out of 100 — highest severity flagged

100%

Incident Confidence

AI classification certainty on critical case

Problem → ThreatLens Solution

THE PROBLEM

→ Alert Overload

Security teams receive alerts from multiple sources and formats simultaneously, with no unified view.

→ Fragmented Investigation

Related events are investigated separately, creating alert noise and costly analyst delays.

→ No Context

Raw alerts lack severity context, attack behavior classification, or recommended response actions.

→ Manual Correlation

Analysts spend hours manually connecting alerts before they can even begin to understand the incident.

THE SOLUTION

→ Multi-Source Ingestion

Normalizes events from SIEM JSON, Threat Feed CSV, and Sensor JSON into a unified schema.

→ Intelligent Correlation

Automatically groups related events into coherent incidents using a correlation engine.

→ Threat Classification

Classifies each cluster as a true threat or false positive, eliminating noise at the source.

→ MITRE ATT&CK Mapping

Maps incidents to ATT&CK techniques with a 0–100 explainable risk score.

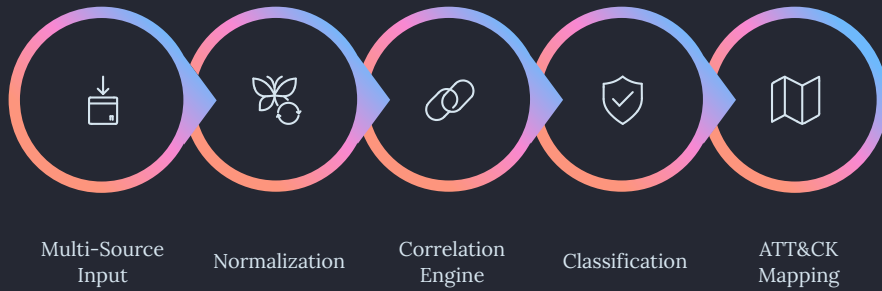
→ BLUF Generation & IBM Bob

Produces commander-level summaries and exposes all intelligence via conversational MCP interface.

📄 **TRANSFORMATION:** RAW ALERTS → THREATLENS → ACTIONABLE INCIDENTS

ThreatLens Architecture & Demo Workflow

PROCESSING PIPELINE



CRITICAL DEMO SCENARIO

Attacker

Source IP: 185.22.14.8 → **Target:** 10.0.0.15

Observed Activities

- Repeated authentication failures
- Network scanning behavior
- Threat-intelligence feed match

ThreatLens Output

- **Risk Score:** 95 / 100
- **Confidence:** 100%
- **Correlated Alerts:** 8 grouped into 1 incident

Input Formats

SIEM JSON · Threat Feed CSV ·
Sensor JSON

Output

Dashboard · BLUF · MITRE Map ·
IBM Bob

IBM Bob + MCP Integration

ThreatLens exposes security intelligence through MCP tools for conversational access via IBM Bob, enabling analysts to investigate incidents in natural language without leaving their workflow.

MCP TOOL CATALOG

Tool	Function
<code>get_priority_incidents()</code>	Find the highest-priority active incidents
<code>get_incident(id)</code>	Retrieve full incident details and evidence
<code>investigate_incident(id)</code>	Deep-dive investigation on a selected incident
<code>get_mitre_mapping(id)</code>	Retrieve MITRE ATT&CK technique mappings
<code>generate_bluf(id)</code>	Generate commander-level BLUF summary

CONVERSATIONAL WORKFLOW

01

Analyst Query

"What is the most dangerous incident right now?"

02

IBM Bob → ThreatLens

Calls `get_priority_incidents()` — returns highest-priority incident with risk score and confidence.

03

Analyst Query

"Investigate incident 1."

04

IBM Bob → ThreatLens

Calls `investigate_incident(1)` — returns full evidence set, risk, confidence, and MITRE techniques.

05

Analyst Query

"Give me a commander-level BLUF."

06

ThreatLens Response

Calls `generate_bluf(1)` — delivers concise summary with risk verdict and recommended actions.

Impact, Results & Future Scope

✓ IMPLEMENTED

- Multi-source event normalization
- Alert correlation engine
- Threat vs. false-positive classification
- Explainable 0–100 risk scoring
- Confidence scoring per incident
- MITRE ATT&CK mapping
- Incident clustering & BLUF generation
- IBM Bob / MCP integration
- Security analyst dashboard

📊 RESULTS

25 → 8

Alerts collapsed into correlated incidents

95/100

Risk score on critical incident

100%

Confidence on critical classification

6

MITRE ATT&CK techniques identified

Reduced analyst investigation time · Eliminated false-positive noise

🔭 FUTURE SCOPE

- Connect additional live SIEM and threat-intelligence sources
- Production-grade persistent storage
- Expand MITRE ATT&CK coverage
- Stronger authentication and authorization
- Continuous monitoring deployment

BLUF EXAMPLE — CRITICAL THREAT DETECTED

"External host **185.22.14.8** launched coordinated credential attack against **10.0.0.15**. Five authentication failures, network scanning, and threat-intelligence match correlated within minutes. MITRE **T1110** and **T1046** indicated. Risk **95/100** with **100% confidence**. Immediately block source IP, investigate authentication logs, check target for compromise."